Modern Microservices & Kubernetes Deployment Guide

Comprehensive Developer Masterclass & Practical Guide

DEVOPS & CLOUD

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Chapter 1: Production Kubernetes Cluster Architecture

Control Plane, Worker Nodes, Networking, and Container Runtimes

Kubernetes abstracts physical infrastructure into a declarative control loop. Mastering Deployments, StatefulSets, DaemonSets, and Cluster Autoscaling is the foundation of high-availability cloud engineering.

Key Takeaways & Best Practices:

- Always specify CPU and Memory requests and limits to prevent noisy neighbor eviction.
- Use Liveness and Readiness probes with sensible initial delays.
- Keep secrets externalized in encrypted vaults or KMS-backed secret managers.

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